

Macroinvertebrate survival during cessation of flow and streambed drying in a lowland stream

RALF C. M. VERDONSCHOT*, AGATA M. VAN OOSTEN-SIEDLECKA*, CAJO J. F. TER BRAAK[†] AND PIET F. M. VERDONSCHOT*[‡]

*Department of Freshwater Ecology, Alterra, Wageningen UR, Wageningen, The Netherlands

[†]Biometris, Wageningen University, Wageningen, The Netherlands

[‡]Institute for Biodiversity and Ecosystem Dynamics, University of Amsterdam, Amsterdam, The Netherlands

SUMMARY

1. The number of perennial low-order lowland streams likely to experience intermittent flow is predicted to increase in north-western Europe. To understand the effects of such a change on macroinvertebrates, a field experiment was carried out in a currently perennial sandy lowland stream.

2. Using a before–after control–impact design, the flow regime was manipulated to yield two distinct treatments: stagnation (although with little water loss) and drying of the stream (although artificial remnant pools remained in the bed). There was also an unmanipulated control reach. The two treatments were applied simultaneously in separate, consecutive reaches, resulting in 29 days of stagnation and 25 days of streambed drying with surface water only present in the remnant pools. Changes in macroinvertebrate richness, abundance and community composition were recorded, and we assessed whether these changes could be explained by ecological preferences for flow of the various taxa.

3. Stagnation resulted in only minor changes in community composition. A small number of rheophilic taxa disappeared, while taxa preferring standing waters complemented those already present, increasing total biodiversity. In remnant pools in the otherwise dry reach, richness and abundance peaked after they became isolated, indicating a concentration of invertebrates. A subsequent steep decline in richness coincided with hypoxia and increasing conductivity. *Culex pipiens/torrentium* colonised the pools and was then dominant. Only a small subset of the assemblage successfully used the dry streambed as a refuge.

4. The effect of a shift from perennial to intermittent flow evidently depends on the degree of habitat change. Environmental conditions after cessation of flow are crucial in determining lowland stream macroinvertebrate persistence during water drawdown.

Keywords: benthic invertebrates, dry sediment, flow permanence, refugia, temporary streams

Introduction

The hydrology of many streams in north-western Europe is strongly affected by human activities. Stream channelisation has increased the speed of runoff and resulted in the loss of natural riparian zones. Concurrently, water abstraction has resulted in an overall decrease in the amount of water supplied to streams; in dry summers, this decrease may result in periods of low flow or streambed drying, especially in headwaters. The

frequency, intensity and duration of such extreme events are expected to increase due to climate change (Verdonschot *et al.*, 2010; KNMI, 2014). These changes will not only affect temporary streams, but will transform the flow regime of many low-order streams from perennial to intermittent. Loss of perennial flow could lead to major changes in stream communities (Bogan & Lytle, 2011).

Low flows and streambed drying can be viewed as ‘ramp disturbances’: situations in which environmental

Correspondence: Ralf C.M. Verdonschot, Department of Freshwater Ecology, Alterra, Wageningen UR, P.O. Box 47, 6700 AA, Wageningen, The Netherlands. E-mail: ralf.verdonschot@wur.nl

conditions steadily get worse as droughts persist (Lake, 2000, 2011). A period of low flow or stagnation, without complete drying of the stream, could result in a number of effects on the macroinvertebrate assemblage due to changed environmental conditions (Stanley *et al.*, 1994; Acuña *et al.*, 2005; Dewson, James & Death, 2007a; Bond, Lake & Arthington, 2008; Boulton & Lake, 2008; Lake, 2011) and an intensification of biotic interactions, such as an increase in intra- and interspecific competition and predation with a decreasing water volume (Covich, Crowl & Scatena, 2003; Stubbington, Wood & Reid, 2011). Streambed drying has especially severe consequences for the biota. Even in intermittent systems, where many organisms have adaptations to resist droughts (Williams, 1987; Lytle & Poff, 2004; Stubbington & Datry, 2013), resilience (resulting from the proximity to permanent water as a source of colonists) plays an important role in community recovery after resumption of flow (Fritz & Dodds, 2004; Acuña *et al.*, 2005; Lake, 2011).

Although macroinvertebrate responses to streambed drying and cessation of flow have been well studied in many temporary lotic systems around the world, knowledge of the macroinvertebrate response to a shift from perennial to intermittent flow is limited (Wood & Petts, 1999; Bogan & Lytle, 2011), partly due to the unpredictability of such events (Lake, 2011). To understand the effects of such a change on macroinvertebrates, it is necessary to quantify the hydrological and physicochemical thresholds for the species present as well as to determine to what extent refuges are used to resist the perturbation. Therefore, we performed a small-scale field experiment in a sandy lowland stream in the Netherlands to examine the effects on macroinvertebrates of a shift from perennial constant flow to episodes of very low flow, resulting in stagnation and streambed drying. Since it is difficult to predict the location and timing of such a transition under natural conditions, the flow regime of the stream was manipulated.

We determined to what extent a stagnant reach, and a dry reach with remnant pools, acted as refuges for stream macroinvertebrates. These treatments represent three successive phases of the perturbation caused by hydrological drought: cessation of flow, habitat reduction and concentration of biota in the pools, and complete loss of surface water. Taxon richness and composition were recorded weekly to determine the direction, timing and magnitude of changes in the macroinvertebrate assemblage. We expected that the effect of the different treatments on the macroinvertebrate assemblages

would be related to the duration and harshness of the treatment applied, with only small and gradual changes in the stagnant reach and larger more rapid changes in the pools and the dry streambed. Furthermore, we expected contrasting responses in survival among taxa, because of differences in sensitivity to changes in environmental conditions. A metric describing preferences for flow was used to differentiate between sensitive and tolerant taxa. We predicted that taxa preferring moderate to fast flow (rheophiles) would be affected more strongly by the treatments applied than taxa more characteristic of standing water.

Methods

Study area

The Hierdense beek is a sandy lowland stream located near Harderwijk, The Netherlands. The stream is approximately 20 km long and up to 3.5 m wide and 80 cm deep. Flow is variable, with daily discharges of 0.04–0.54 m³ s⁻¹ and current velocities of 8–50 cm s⁻¹. Periods of very low flow are rare in the middle and lower courses of the stream, while drying of the stream was last reported during the extreme drought of summer 1976. The middle course of the stream runs through a forested area, the Leuvenumse bos, where it consists of a number of anastomosing channels; in one of which, the experiments were carried out (latitude 52°20'10.69" N, longitude 5°42'47.52"E). This channel (200 m long, on average 2 m wide and 0.2 m deep) was fully shaded by mixed deciduous forest, and the substratum was dominated by sand, coarse particulate organic material (CPOM) and fine particulate organic matter (FPOM) with small patches of gravel, loam, wood and emergent macrophytes.

Experimental design

We used a before–after control–impact design to assess changes in macroinvertebrate assemblage composition following periods of low flow and streambed drying. The flow regime of the stream channel was manipulated to cause two distinct treatments: loss of flow (stagnation) and drying. The treatments were applied simultaneously in separate, consecutive reaches of the stream channel and compared to an unaffected control reach (Fig. 1). The most downstream reach was used for the streambed drying treatment, and the stagnation treatment was applied upstream of this reach. A reach further

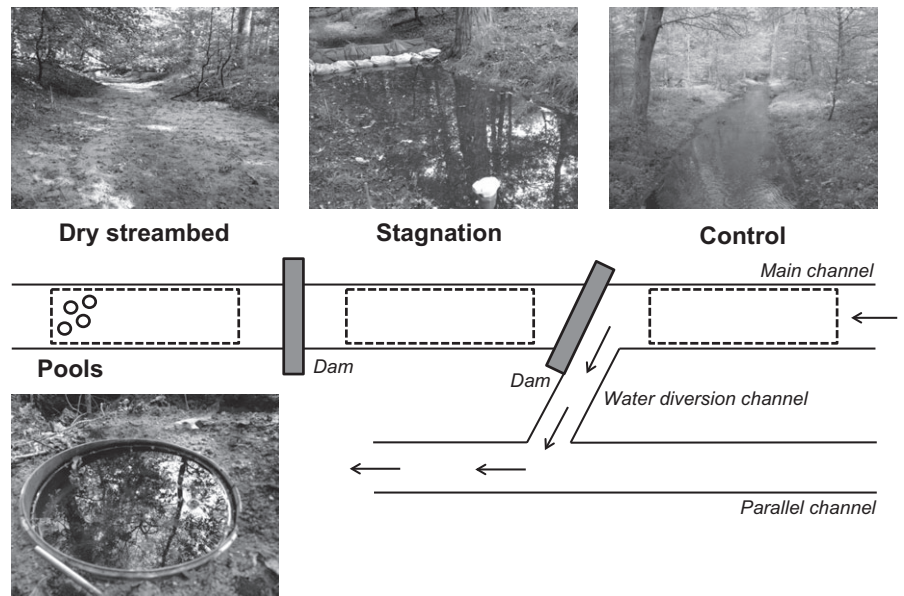


Fig. 1 Outline of the experimental set-up in the Hierdense beek (not to scale). Dashed lines indicate reach positions.

upstream was used as the control. Each reach was approximately 30 m long. In the dry reach, 15 white buckets (diameter 25 cm, height 17.5 cm) were embedded into the streambed prior to the manipulation of flow regime to mimic the formation of small, isolated, remnant pools during water drawdown (Fig. 1). Bed sediments excavated within the reach were placed in each bucket to a depth of approximately 7.5 cm, resulting in a mean water depth of 10 cm (± 1 cm) when the pools were formed.

Manipulation of the flow regime required the construction of two dams in the stream channel, one between the dry and stagnant reaches and one between the stagnant and control reaches (Fig. 1). The dams were approximately 0.3 m high and consisted of high-pressure laminate plates sealed with plastic sheeting. Sandbags were used to strengthen the construction and to keep the plastic sheeting in place. Both dams were constructed on 9 July 2012. To divert the water flowing from the control reach, a diversion channel was dug and connected to a nearby parallel channel, creating a bypass. Due to very high discharge, water continued to flow for another 2 weeks through the main channel before the diversion became effective, with onset of stagnation occurring on 22 July 2012. Water loss in the stagnant reach was minimal; the water volume generally followed that of the control reach (Table 1). This hydrological linkage between the control and stagnant reaches was the result of water slowly seeping through or under the dam. Given the fine grain size of the streambed and the plastic sheeting covering the bed upstream of the dam, we do not regard this hydrological connection to

be a significant pathway for stream invertebrates entering the stagnant reach from the control reach. On 26 July 2012, there was no surface water left in the dry reach, resulting in complete isolation of the remnant pools.

Environmental variables

To investigate changes in conditions in the stagnant reach after the loss of flow, in comparison to the control reach, we measured dissolved oxygen (DO) saturation, temperature, pH, conductivity, nutrients and assessed the substratum. Parameters were recorded at 1-week intervals between 12 June 2012 and 20 August 2012 (five samples before manipulation and five samples after manipulation). The first post-manipulation sample was taken 1 day after the onset of stagnation and the formation of remnant pools. To measure DO saturation and water temperature, Hach LDO dissolved oxygen probes (Hach Company, Loveland, CO, USA) were used. Conductivity and pH were directly measured in the field with electrodes (Hach Company), and 1 L water samples were taken for laboratory analysis of nutrients (total phosphorus, ammonium, nitrate and nitrite). No water samples were taken from the remnant pools due to their small water volumes. Substratum composition was estimated visually in one fixed 5-m longitudinal transect per reach, and the substratum types observed [sand, fine particulate organic matter (FPOM), coarse particulate organic matter (CPOM), rooted macrophytes and filamentous algae] were expressed on a percentage cover scale (0–100%). The moisture condition of the surficial sediment of the dry streambed was estimated visually as

Table 1 Measurements of stream width, depth and velocity pre- and post-manipulation to illustrate the effect of the experimental treatment on the streams hydrology. Variables were measured in one fixed 5-m longitudinal transect per reach

Date (day/month)		18/06	26/06	4/07	9/07	23/07	30/07	6/08	13/08	20/08
Stagnation (days pre/post)		-34	-26	-18	-13	1	8	15	22	29
Loss of surface water (days pre/post)		-37	-29	-21	-16	-3	4	11	18	25
Reach width (m)	Control	1.5	2.0	1.5	2.5	3.0	2.5	2.0	2.0	0.5
	Stagnation	2.0	2.5	2.0	3.0	3.0	2.5	2.0	2.5	0.5
	Dry bed	2.0	2.5	2.5	3.5	2.0	–	–	–	–
Reach depth _{max} (m)	Control	0.13	0.15	0.10	0.21	0.26	0.18	0.19	0.16	0.05
	Stagnation	0.08	0.13	0.09	0.19	0.26	0.18	0.10	0.18	0.06
	Dry bed	0.12	0.12	0.06	0.17	0.04	–	–	–	–
V _{max} (m s ⁻¹)	Control	0.19	0.31	0.10	0.37	0.21	0.16	0.14	0.12	0.06
	Stagnation	0.23	0.33	0.18	0.41	0.00	0.00	0.00	0.00	0.00
	Dry bed	0.19	0.31	0.16	0.37	0.09	–	–	–	–
Q reach (m ³ s ⁻¹)	Control	0.03	0.06	0.01	0.11	0.08	0.05	0.03	0.02	<0.01
	Stagnation	0.03	0.05	0.01	0.11	–	–	–	–	–
	Dry bed	0.03	0.06	0.01	0.10	<0.01	–	–	–	–

dry (dusty, dry to the touch), moist (damp but no visible water) or wet (visible free water).

Oligochaeta, which were omitted because of identification difficulties (especially non-mature individuals).

Macroinvertebrates

The composition of the invertebrate assemblages in the reaches was assessed from samples taken once per week, five times before and five times after manipulation of the flow regime. The first invertebrate sample was taken on 12 June 2012 and the last on 20 August 2012. To avoid repeatedly sampling the same locations in the reaches, each reach was divided into 20 1.5-m sections; one section was chosen randomly for each sampling occasion, omitting previously sampled sections.

Three sampling methods were used. Reaches containing water (all experimental reaches before manipulation, the stagnant reach after manipulation and the control reach) were sampled using a Surber sampler (25 × 25 cm; 0.5 mm mesh). In total, six Surber sample units were taken per sampling occasion from the dominant substratum present in that section. The dry streambed was sampled in a similar manner, but instead of using the Surber sampler, six sediment samples were collected (surface area 25 × 25 cm, excavated up to 5 cm deep) (Stubington *et al.*, 2009). Finally, the fauna of three randomly chosen remnant pools was collected. Buckets were sealed first to prevent leakage and loss of macroinvertebrates and then excavated from the streambed. All samples were transported to the laboratory, sieved over 1-mm and 250-µm mesh and sorted for live organisms. Macroinvertebrates were identified to the lowest practical taxonomic level, except for the

Data analysis

Post-manipulation changes in environmental parameters were investigated with repeated measures with autoregressive error (AR1) using GenStat for Windows 16th edition (VSN International, Hemel Hempstead, UK) and randomised intervention analysis (RIA) (Carpenter *et al.*, 1989) using Canoco 5.03 (Ter Braak & Šmilauer, 2012). The null hypotheses of no change in the stagnant reach or pools relative to the control reach were evaluated for each environmental variable against the alternative of time-dependent changes (five degrees of freedom). Significance in the RIA was calculated with 9999 Monte Carlo random permutations restricted for the temporal structure of the data (cyclic shifts). Two different versions of RIA were applied. In the first version, the same cyclic shift of the time series was applied to the stream sections involved, whereas they were independently randomly shifted in the second version. The first version is arguably most appropriate for random intervention analysis but is very conservative; as there are just 10 time points, its lowest attainable *P*-value is 0.10. The second version is more similar to the repeated measures model but makes no assumption on the order of the autocorrelation.

To investigate how manipulation of the flow regime influenced macroinvertebrate composition, changes in taxon richness, abundance and assemblage composition were examined. To facilitate the simultaneous analysis of the four conditions (control, stagnation, streambed

drying and remnant pools), three of the six replicates of the reach used for the streambed drying treatment taken at each of the five pre-manipulation times were assigned to the streambed drying treatment and the other three to the remnant pools. Thereafter, the data were averaged for each condition–time combination, and the same statistical analyses were performed as for the environmental variables, but now with four conditions instead of just two. Assemblage composition could not be analysed with repeated measures.

Non-metric multidimensional scaling (NMDS) was used to display changes in the macroinvertebrate assemblages of the reaches over time. As before, samples were averaged for this analysis, and taxon abundances were $\log_2(x + 1)$ -transformed to reduce the influence of dominant taxa in the data set. Non-metric multidimensional scaling was based on a Bray–Curtis dissimilarity matrix and was conducted in PC-ORD for Windows version 4.25 (slow and thorough autopilot mode, 400 iterations, instability criterion of 0.00001, starting number of axes of 6, 40 real runs and 50 runs with randomised data; McCune & Grace, 2002). In addition, we applied principal response curve (PRC) analysis (van den Brink & ter Braak, 1999) to reveal which taxa changed after manipulation. Principal response curve analysis is used here to show the time-dependent condition effects on an assemblage of taxa, expressed as differences from the control condition. For this analysis, the original six (or three) replicates per time point were divided in two sets and then averaged per set, giving two replicates per condition–time combination. This reduced the amount of pseudo-replication while still allowing a graphical display of the post-manipulation differences among the conditions as compared to the pre-manipulation conditions. Principal response curve analysis was carried out using Canoco 5.03 (Ter Braak & Šmilauer, 2012).

Initial survival was determined by establishing whether the taxa recorded in the reaches before manipulation were recorded after manipulation. The next step was to determine which taxa persisted, indicating the refuge value of the different treatments applied to the reaches. If a taxon failed to survive the entire treatment period, survival was defined as poor. Taxa which appeared in the manipulated reaches but were not in the control reach and/or in the manipulated reaches before the start of the manipulation were regarded as newcomers. To assess whether rheophilic taxa were affected more strongly by the treatments than the rest of the community, a metric describing the current preference of European macroinvertebrates was used (Schmedtje & Colling, 1996; AQEM consortium, 2002).

The metric distinguishes between seven current velocity preference types, ranging from taxa occurring only in standing waters (type 'limnobiont') to taxa restricted to zones with high current ('rheobiont'). Only the taxa of two types are restricted to streams, 'rheobiont' and 'rheophilic' (taxa of zones with moderate to high current). Current preferences of the taxa recorded during the last sampling occasion were derived from the freshwater ecology.info-database (Schmidt-Kloiber & Hering, 2012). For each sample, both the percentage of the total numerical abundance and the number of taxa classified as 'rheophilic' or 'rheobiont' were calculated. Difference in relative abundance between treatments was compared using a Kruskal–Wallis nonparametric one-way ANOVA test, with Mann–Whitney U-tests to examine differences between treatment pairs. The mean number of taxa recorded in the different treatments was compared using a one-way ANOVA followed by Tukey's *post hoc* procedures. Statistical tests were conducted in IBM SPSS Statistics version 19.0 (IBM Corp, Armonk, NY, USA).

Results

Physicochemical conditions and substratum composition

Stagnation resulted in a small decrease in oxygen saturation post-manipulation in comparison to the control ($F_{5,9} = 18.7$, $P < 0.01$; Tables 2 and 3) and resulted in a marked decrease in the percentage of the substratum that consisted of sand ($F_{5,9} = 14.3$, $P < 0.01$; Table 3, Fig. 2). No significant effects on nutrient concentrations, conductivity, pH or water temperature were detected. Oxygen saturation in the remnant pools was lower post-manipulation than in the control ($F_{5,9} = 426.8$, $P < 0.05$), this decline being apparent by day 4, when saturation values had declined to 16%, and remained low (0–35%) subsequently (Table 2, 3). Conductivity in the pools differed from the control post-manipulation ($F_{5,9} = 45.7$, $P < 0.05$), first increasing from 312 to 505 $\mu\text{S cm}^{-1}$ but then declining to 237 $\mu\text{S cm}^{-1}$ at day 25. Conductivity in the control reach remained between 314 and 335 $\mu\text{S cm}^{-1}$ throughout. No significant differences in pH or temperature were observed in comparison to the control. The moisture content of the surficial sediment of the dry streambed decreased during the experiment from wet (day 1) to moist (days 4–18) to dry (day 25).

Macroinvertebrate assemblage richness and composition

Before manipulations began, 64 taxa were collected (8024 individuals). The most abundant taxa, representing

Table 2 Physicochemical measurements before and after flow manipulation

Time (days) pre/post	Treatment	Variable						
		Tot-P (mg L ⁻¹)	NH4 (mg L ⁻¹)	NO3 + NO2 (mg L ⁻¹)	EC (µS cm ⁻¹)	pH	T (°C)	DO (%)
-40	Control	0.04	0.01	5.17	326	7.5	14.0	91
-34	Control	0.01	0.03	4.85	312	7.4	14.3	87
-26	Control	0.07	0.03	4.61	321	7.5	12.8	87
-18	Control	0.03	0.01	5.01	329	7.5	17.3	90
-3	Control	0.09	0.11	3.73	291	7.5	15.4	81
1	Control	0.06	0.04	4.66	314	6.8	14.2	88
8	Control	0.06	0.03	4.70	324	6.9	14.8	94
15	Control	0.05	0.03	4.25	335	6.8	16.6	99
22	Control	0.04	0.02	5.03	331	7.6	15.5	91
29	Control	0.05	0.05	4.91	329	6.9	21.4	84
-40	Stagnation	0.05	0.04	5.14	329	7.5	13.9	92
-34	Stagnation	0.02	0.03	4.84	317	7.5	14.3	89
-26	Stagnation	0.06	0.04	4.52	317	7.4	13.3	90
-18	Stagnation	0.04	0.03	5.03	331	7.5	17.4	92
-13	Stagnation	0.1	0.13	3.65	292	7.3	15.4	84
1	Stagnation	0.07	0.04	4.47	314	7.0	15.0	84
8	Stagnation	0.05	0.04	4.73	321	7.1	15.1	85
15	Stagnation	0.06	0.03	4.91	325	6.8	16.8	93
22	Stagnation	0.05	0.03	4.96	329	6.8	15.1	87
29	Stagnation	0.04	0.08	4.87	280	7.3	24.7	56
-37	Pools	–	–	–	312	7.5	12.7	93
-29	Pools	–	–	–	317	7.4	14.4	87
-21	Pools	–	–	–	318	7.5	13.0	86
-16	Pools	–	–	–	329	7.5	17.3	90
-3	Pools	–	–	–	288	7.5	15.3	82
1	Pools	–	–	–	312	6.8	15.8	87
4	Pools	–	–	–	420	6.9	15.5	16
11	Pools	–	–	–	449	6.8	16.8	0
18	Pools	–	–	–	505	7.6	15.1	35
25	Pools	–	–	–	237	6.9	20.8	20

>75% of the total abundance, were *Stempellinella brevis* (Diptera; 22.7%), *Gammarus pulex* (Amphipoda; 21.9%), *Nemoura cinerea/avicularis* (Plecoptera; 8.2%), *Hydropsyche pellucidula* (Trichoptera; 7.7%), *Micropsectra* sp. (Diptera; 6.6%), *Baetis vernus* (Ephemeroptera; 4.0%), Ceratopogonidae (Diptera; 3.0%) and *Cladotanytarsus* sp. (Diptera; 2.6%). In the weeks after the treatments were applied, 61 taxa (3760 individuals) were found the control reach, 79 taxa (4918 individuals) in the stagnant reach, 47 taxa (4979 individuals) in the pools and 41 taxa (1288 individuals) were recovered from the dry streambed.

Post-manipulation patterns in richness and abundance in the manipulated reaches differed from that of the control reach (Table 4). The number of taxa and invertebrate density in the control and stagnant reach increased in the weeks following manipulation of the flow regime (Fig 3a,b), although in the stagnant reach, a temporal drop in richness to pre-manipulation value was observed at day 15 after manipulation. Remnant pool

formation initially resulted in detection of many taxa and individuals in the samples, followed by a strong decrease (Fig. 3a,b). In the following weeks, richness decreased further while invertebrate density increased markedly. Streambed drying resulted in a decrease in richness and invertebrate density in the surficial sediment (Fig. 3a,b).

Manipulation of the flow regime affected the taxonomic composition of experimental reaches (Table 4). Non-metric multidimensional scaling of the macroinvertebrate assemblages resulted in an optimal solution incorporating two axes (final stress for two-dimensional solution, 0.13) (Fig. 4). In terms of taxonomic composition, the control and stagnant reaches were comparable. In the weeks after streambed drying and pool formation, marked changes in assemblage composition occurred: dissimilarity increased among samples, as well as between the dry streambed and the pool samples. The PRC diagram of the macroinvertebrate assemblages of

Table 3 Before–after control–impact (BACI) analyses on physico-chemical measurements and substratum cover estimates using repeated measures with an AR1 error model ($F_{5,9}$, $P_{\text{rep. meas.}}$) for all variables and two permutation tests (9999 permutations) based on independent and dependent random shifts of the time series (P_{shift1} and P_{shift2} , respectively) for selected variables with low $P_{\text{rep. meas.}}$

Variable	n^*	$F_{5,9}^\dagger$	$P_{\text{rep. meas.}}$	P_{shift1}	P_{shift2}
Tot-P (stagnation)	10	0.1	0.98		
NH ₄ (stagnation)	10	0.5	0.75		
NO ₃ + NO ₂ (stagnation)	10	0.7	0.61		
EC (stagnation)	10	1.9	0.19	0.12	0.30
EC (pools)	10	45.7	<0.001	0.03	0.10
pH (stagnation)	10	1.0	0.47		
pH (pools)	10	0.4	0.81		
DO (stagnation)	10	18.7	<0.001	0.005	0.10
DO (pools)	10	426.8	<0.001	0.037	0.20
T (stagnation)	10	2.3	0.15		
T (pools)	10	0.8	0.56		
Sand (stagnation)	9	14.3	<0.001	0.007	0.10
FPOM (stagnation)	9	2.4	0.12		
CPOM (stagnation)	9	0.3	0.90		
Filamentous algae (stagnation)	9	0.3	0.90		
Rooted macrophytes (stagnation)	9	0.1	0.98		

*Number of paired samples.

† Approximate denominator degrees of freedom varied between 7.1 and 10.7. All variables were log-transformed before analysis.

all treatments showed that the assemblages of the remnant pools and the dry streambed clearly differed from the control in the weeks following treatment, while the taxonomic composition of the stagnant reach remained similar to the control (PRC-axis 1, 35.2% of variation explained by treatment; Monte Carlo, $P < 0.01$) (Fig. 5a). This difference was the result of many taxa decreasing in the weeks after manipulation in the pools and dry streambed, as indicated by the large number of positive taxon weights in the PRC. Only *Culex pipiens/torrentium* had a relatively high negative weight and thus was increasing following treatment (Fig. 5b). The differences in assemblage composition between the remnant pools and the dry streambed, as displayed in the NMDS diagram (Fig. 4), were captured along PRC-axis 2 (15.9% of variation explained by treatment; Monte Carlo, $P < 0.01$). We visualised this difference by a PRC on the dry streambed and pools treatments only (Fig. 5c). The presence of *Gammarus pulex*, *Culex pipiens/torrentium*, *Asellus aquaticus*, *Glossiphonia complanata*, *Zaorelimyia* sp., *Chironomus* sp. and *Erpodella octoculata* differentiated the pools from the dry streambed, which was characterised by relatively high negative taxon weights for Ceratopogonidae, Limoniidae (except *Dicranota*) and Rhagionidae (Fig. 5d).

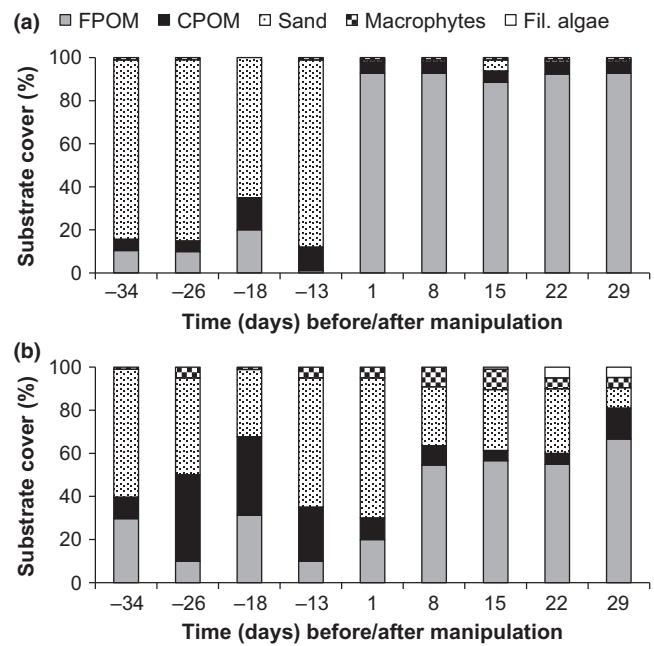


Fig. 2 Substratum cover (visual estimate of 5-m longitudinal transects) on a percentage cover scale (0–100%) in the stagnant (a) and control (b) reaches before and after manipulation of the flow regime. FPOM, fine particulate organic matter; CPOM, coarse particulate organic matter; fil., filamentous algae.

Macroinvertebrate survival and colonisation after manipulation

Although a high percentage of the macroinvertebrate assemblage was initially recovered (of the 64 taxa recorded before treatment, 88% were recovered in the stagnant reach, 63% in the pools and 56% in the sediment of the dry streambed, with 29 taxa appearing in all treatments), many taxa disappeared in the weeks following manipulation. Several groups were distinguished based on survival (Table 5). Only three taxa survived

Table 4 Before–after control–impact (BACI) analyses on mean number of macroinvertebrate taxa and individuals and on the taxon community (abundances) by repeated measures with an AR1 error model ($F_{5,9}$, $P_{\text{rep. meas.}}$) for the first two variables and two permutation tests (9999 permutations) based on independent and dependent random shifts of the time series (P_{shift1} and P_{shift2} , respectively) for all variables

Variable	n^*	$F_{15,15}^\dagger$	$P_{\text{rep. meas.}}$	P_{shift1}	P_{shift2}
Mean no of taxa	40	4.6	0.002	0.013	0.10
Mean abundance	40	5.7	<0.001	0.090	0.20
Community	40	7.0	NA	0.020	0.10

*Number of paired samples.

† Approximate denominator degrees of freedom. All variables were log-transformed before analysis.

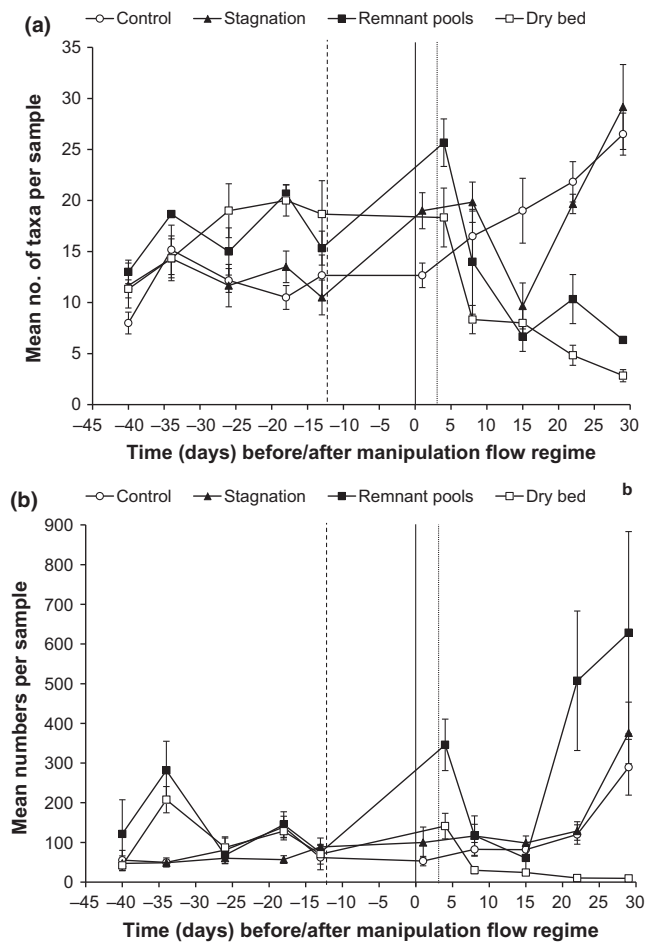


Fig. 3 Mean ($\pm$ SE) number of macroinvertebrate taxa (a) and individuals (b) in each sample (control and stagnation, $n = 6$; dry bed and pools, $n = 3$) for each sampling occasion before and after manipulation of the flow regime. Dashed line, dam construction; solid line, dam effective, start of treatments; dotted line, no surface water left in dry reach.

the entire experimental period in all treatments applied: *Pisidium* sp., *Sialis* sp. and Ceratopogonidae. In total, seven taxa survived singly in the control reach, of which especially the common taxa *Hydropsyche pellicidula*, *Tvetenia* sp. and *Dicranota* sp. displayed contrasting temporal patterns among treatments. Most taxa (31) survived in both the control and the stagnant reach, but were lost from the sediment of the dry streambed or the remnant pools. All Ephemeroptera, Plecoptera and most Trichoptera belonged to this group, as well as many Diptera and Hydracarina. A small, but distinctive group (five taxa) was formed by the taxa which survived in the control and stagnant reaches plus remnant pools in the dry reach: all Hirudinea and Crustacea managed to survive in both the pools and the stagnant reach. Only *Dryops* sp. beetle larvae survived in the sediment of the dry

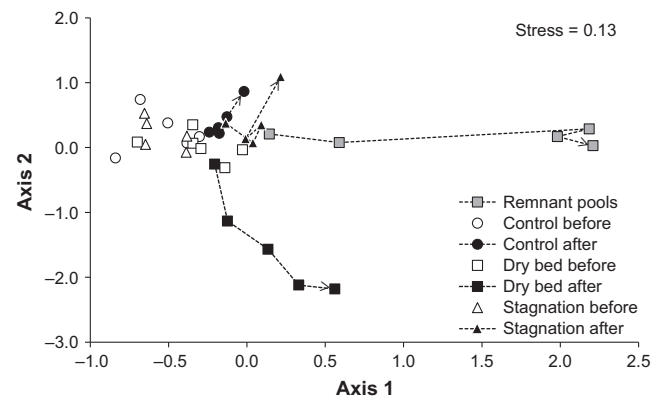


Fig. 4 Non-metric multidimensional scaling plot of sites based on Bray–Curtis dissimilarities. Samples taken after manipulation are connected by dotted lines, with an arrow pointing to the last sample taken after manipulation.

streambed; *Chrysops* sp. larvae were also found in the stagnant reach. Another 17 taxa were found before and after manipulation of the flow regime, but were either absent in the control reach or were not found in the control reach on the last sampling occasion.

After manipulation, 18 taxa appeared in the experimental reaches which were not found in the control reach and/or before the manipulation started (Table 6). In particular, the stagnant reach contained many taxa not found elsewhere. Remarkably, in the stagnation treatment, all newcomers were found in samples taken on the last sampling occasion, while colonisation of the remnant pools had occurred by day 11 after they were isolated. Most new taxa were not abundant in the experimental reaches, with the mass colonisation of the pools by *Culex pipiens/torrentium* and, to a lesser extent, *Chironomus* sp. being exceptions. The newcomers observed in the sediment of the dry streambed (larvae of Ephydriidae and Muscidae) could potentially be terrestrial, since more precise identification was not possible. Most newcomers preferred lentic conditions (Table 6).

Differences in survival among taxa based on flow preference

In total, 11 rheophilic taxa were recorded in the study stream (12% of the total number of taxa); rheobiont taxa were absent. Both proportional abundance (Kruskal–Wallis test, $\chi^2 = 13.8$, $P < 0.001$; Fig. 6a) and the number of rheophilic taxa (One-Way ANOVA, $F_{2,14} = 19.4$, $P < 0.001$; Fig. 6b) differed between treatments. Relative abundance of rheophiles was higher in the control and stagnation reaches than in the sediment of the dry streambed (Mann–Whitney U, $P = 0.002$). The number of

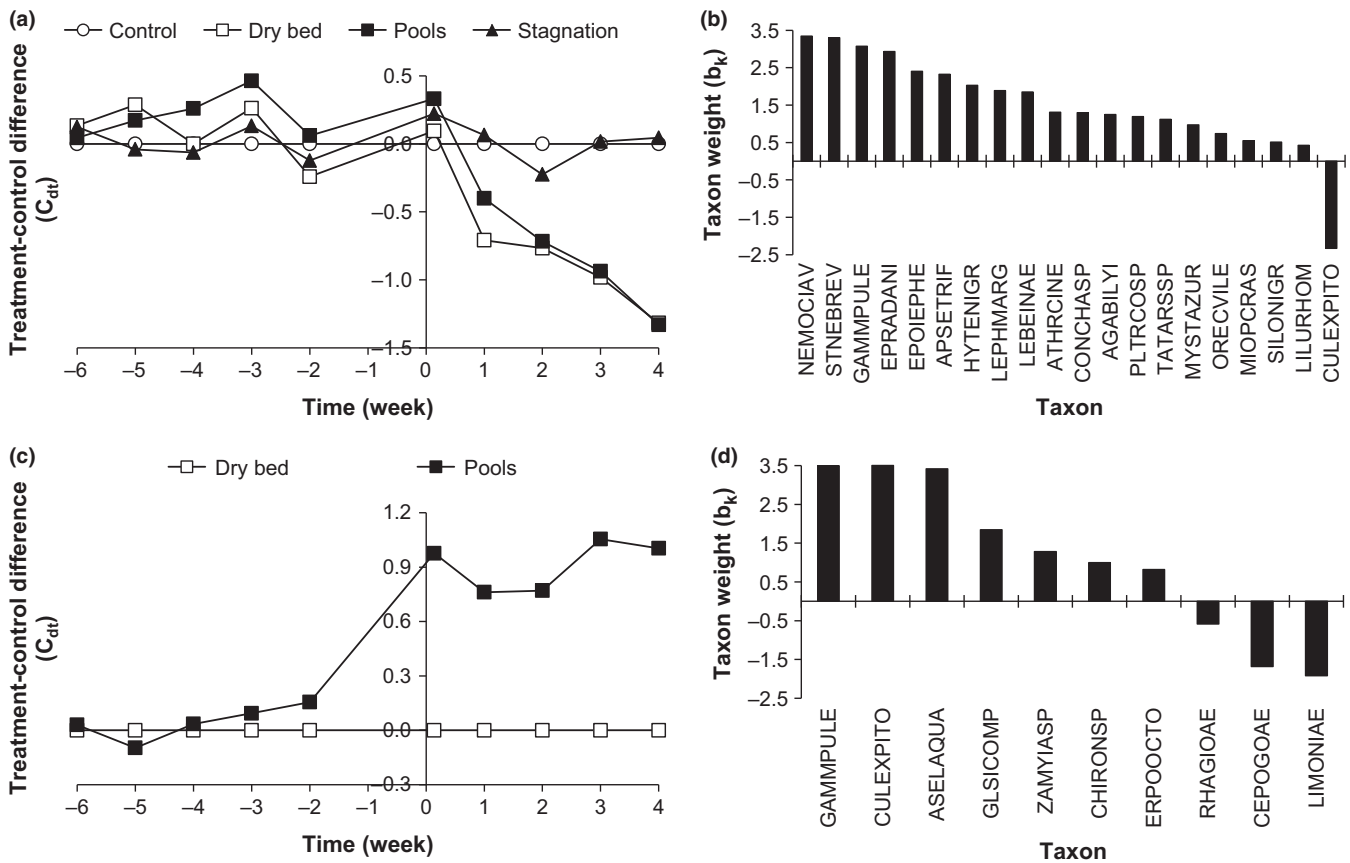


Fig. 5 Principal response curves (C_{dt}) of treatment effects on the macroinvertebrate community for all treatments (a) and the dry streambed and remnant pools only (c). The lines represent the effect of the treatments along time. Species weights (b_k) show the relative contribution of individual taxa to the community response for all treatments (b, limited to 20 taxa by fit) and dry bed-pools (d, limited to 10 taxa by fit). AGABILIYI: *Agabus/ilybius* larva, APSETRIF: *Apsectrotanypus trifascipennis*, ASELAQUA: *Asellus aquaticus*, ATHRCINE: *Athripsodes cinereus*, CEPOGOAE: Ceratopogonidae, CHIRONSP: *Chironomus* sp., CONCHASP: *Conchapelopia* sp., CULEXPITO: *Culex pipiens/torrentium*, EPOIEPHE: *Epoicocladus ephemerae*, EPRADANI: *Ephemera danica*, ERPOOCTO: *Erpobdella octoculata*, GAMMPULE: *Gammarus pulex*, GLSICOMP: *Glossiphonia complanata*, HYTENIGR: *Hygrobates nigromaculatus*, LEBEINAE: *Lebertia inaequalis*, LEPHMARG: *Leptophlebia marginata*, LILURHOM: *Limnephilus rhombicus*, LIMONIAE: Limoniidae except *Dicranota*, MIOPCRAS: *Mideopsis crassipes*, MYSTAZUR: *Mystacides azurea*, NEMOCIAV: *Nemoura cinerea/avicularis*, ORECVILE: *Orectochilus villosus* larva, PLTRCOSP: *Plectrocnemia conspersa*, RHAGIOAE: Rhagionidae, SILONIGR: *Silo nigricornis*, STNEBREV: *Stempellinella brevis*, TATARSSP: *Tanytarsus* sp., ZAMYIASP: *Zavrelimyia* sp.

rheophilic taxa was highest in the control reach, followed by the stagnant reach. In the dry streambed samples, rheophilic taxa were absent while, in the pools, only one taxon was found.

Discussion

Effects of stagnation

Most taxa remained in the stagnant reach throughout the 29-day study period. This is in line with the effects of stagnation described for other lowland streams and rivers, in which the majority of the species present were not adversely affected (Moth Iversen *et al.*, 1978; Extence, 1981). Contrastingly, in other stream systems,

major decreases in richness and density were observed after flow ceased (Miller & Golladay, 1996; Boulton, 2003). The loss of rheophiles is particularly frequently observed when current velocity decreases (Boulton, 2003; Dewson *et al.*, 2007a; Bond *et al.*, 2008; Boulton & Lake, 2008). Rheophiles are sensitive because of the requirement for flow for respiration or feeding (Hynes, 1970). A similar response was observed in our study; the number of rheophilic taxa was lower after flow ceased than in the control. Nonetheless, the number of taxa affected was relatively low in comparison to total richness, which explains the minor community changes overall.

One of the reasons for the disparity in community response among stream systems lies in the widely

Table 5 Temporal patterns of post-manipulation macroinvertebrate survival for the different treatments: $t_{\max}$ indicates days from the time from the start of stagnation/streambed drying and pool formation ($t = 1$, day 1/1; $t = 2$, day 8/4; $t = 3$, day 15/11; $t = 4$, day 22/18; $t = 5$, day 29/25) a taxon was recorded, frequency (f) indicates the number of sampling occasions a taxon was found. Presence is based on the summed density in all replicates per treatment and sampling occasion; relatively common taxa (total numbers ≥ 50 individuals) are marked with an asterisk

Taxon	Taxonomic group	C		D		P		S	
		<i>t</i> _{max}	f (#)	<i>t</i> _{max}	f (#)	<i>t</i> _{max}	f (#)	<i>t</i> _{max}	f (#)
Control									
<i>Hydropsyche pellucidula</i> *	Hydropsychidae (TR)	5	5	1	1	1	1	1	1
<i>Lype reducta/phaeopa</i>	Psychomyiidae (TR)	5	3	1	1	–	–	1	1
<i>Limnius volckmari</i> larva	Elmidae (CO)	5	2	–	–	–	–	–	–
<i>Dicranota</i> sp.*	Limoniidae (DI)	5	5	4	3	1	1	4	4
<i>Brillia bifida</i>	Orthocladiinae (DI)	5	2	2	2	–	–	1	1
<i>Tvetenia</i> sp.*	Orthocladiinae (DI)	5	5	–	–	–	–	1	1
<i>Cricotopus</i> sp.	Orthocladiinae (DI)	5	1	–	–	2	1	–	–
Control, stagnant									
<i>Hygrobatas fluviatilis</i>	Hygrobatidae (HYC)	5	5	–	–	–	–	5	1
<i>Hygrobatas nigromaculatus</i> *	Hygrobatidae (HYC)	5	5	–	–	2	2	5	5
<i>Lebertia inaequalis</i> *	Lebertiidae (HYC)	5	5	1	1	2	2	5	5
<i>Mideopsis crassipes</i>	Mideopsidae (HYC)	5	3	3	2	2	1	5	4
<i>Sperchon setiger</i> *	Sperchontidae (HYC)	5	5	1	1	–	–	5	4
<i>Ephemera danica</i> *	Ephemeridae (EP)	5	5	3	3	2	2	5	5
<i>Baetis vernus</i> *	Baetidae (EP)	5	5	1	1	1	1	5	3
<i>Proclleon bifidum</i>	Baetidae (EP)	5	1	–	–	–	–	5	1
<i>Leptophlebia marginata</i> *	Leptophlebiidae (EP)	5	5	1	1	4	4	5	5
<i>Nemoura cinerea/avicularis</i> *	Nemouridae (PL)	5	5	3	3	4	3	5	5
<i>Agabus/Ilybius</i> larva*	Dytiscidae (CO)	5	4	1	1	4	3	5	5
<i>Orectochilus villosus</i> larva	Gyrinidae (CO)	5	3	–	–	–	–	5	2
<i>Polypedilum</i> sp.*	Chironomini (DI)	5	5	4	4	2	2	5	5
<i>Epoicocladus ephemeræ</i> *	Orthocladiinae (DI)	5	5	2	2	2	2	5	5
<i>Heterotrissocladius marcidus</i> *	Orthocladiinae (DI)	5	5	3	3	1	1	5	5
<i>Corynoneura</i> sp.	Orthocladiinae (DI)	5	2	–	–	2	1	5	3
<i>Stempellinella brevis</i> *	Tanytarsini (DI)	5	5	1	1	1	1	5	5
<i>Micropsectra</i> sp.*	Tanytarsini (DI)	5	5	1	1	–	–	5	5
<i>Tanytarsus</i> sp.*	Tanytarsini (DI)	5	5	–	–	1	1	5	4
<i>Apsectrotanypus trifascipennis</i> *	Tanypodinae (DI)	5	5	2	2	4	3	5	5
<i>Conchapelopia</i> sp.*	Tanypodinae (DI)	5	5	1	1	4	3	5	5
<i>Zavrelimyia</i> sp.*	Tanypodinae (DI)	5	4	–	–	4	4	5	3
<i>Mystacides azurea</i>	Leptoceridae (TR)	5	3	–	–	–	–	5	5
<i>Athripsodes cinereus</i> *	Leptoceridae (TR)	5	3	1	1	4	4	5	2
<i>Plectrocnemia conspersa</i> *	Polycentropodidae (TR)	5	4	–	–	–	–	5	3
<i>Polycentropus irroratus</i>	Polycentropodidae (TR)	5	2	–	–	–	–	5	3
<i>Silo nigricornis</i>	Goeridae (TR)	5	2	–	–	–	–	5	1
<i>Anabolia nervosa</i>	Limnephilidae (TR)	5	1	–	–	2	1	5	2
<i>Halesus radiatus</i>	Limnephilidae (TR)	5	3	–	–	2	2	5	3
<i>Limnephilus rhombicus</i>	Limnephilidae (TR)	5	1	–	–	–	–	5	2
<i>Micropterna sequax</i>	Limnephilidae (TR)	5	1	1	1	–	–	5	2
Control, pool, stagnant									
<i>Erpobdella octoculata</i>	Erpobdellidae (HI)	5	1	-	-	5	4	5	2
<i>Glossiphonia complanata</i> *	Glossiphoniidae (HI)	5	3	-	-	5	5	5	3
<i>Asellus aquaticus</i> *	Asellidae (IS)	5	5	1	1	5	5	5	5
<i>Gammarus pulex</i> *	Gammaridae (AM)	5	5	1	1	5	5	5	5
<i>Empididae</i> *	Empididae (DI)	5	5	4	3	5	3	5	5
Control, dry bed									
<i>Dryops</i> sp. larva	Dryopidae (CO)	5	4	5	5	2	1	4	2
Control, dry bed, stagnant									
<i>Chrysops</i> sp.	Tabanidae (DI)	5	5	5	5	3	2	5	4
All treatments									
<i>Psidium</i> sp.*	Sphaeriidae (BI)	5	5	5	5	5	5	5	5

Table 5 (Continued)

Taxon	Taxonomic group	C		D		P		S	
		$t_{\max}$	f (#)	$t_{\max}$	f (#)	$t_{\max}$	f (#)	$t_{\max}$	f (#)
<i>Sialis</i> sp.*	Sialidae (ME)	5	4	5	5	5	5	5	5
Ceratopogonidae*	Ceratopogonidae (DI)	5	5	5	5	5	3	5	5
Others									
<i>Oxus setosus</i>	Oxidae (HYC)	–	–	–	–	–	–	5	4
<i>Wettina podagrica</i>	Wettinidae (HYC)	–	–	–	–	1	1	5	1
<i>Ilybius fuliginosus</i> adult	Dytiscidae (CO)	–	–	–	–	–	–	5	2
<i>Dixella</i> sp.	Dixidae (DI)	–	–	–	–	–	–	5	2
<i>Phaenopsectra</i> sp.	Chironomini (DI)	–	–	–	–	–	–	5	3
<i>Macropelopia</i> sp.*	Tanypodinae (DI)	3	3	5	4	2	2	5	4
Limoniidae except <i>Dicranota</i> *	Limoniidae (DI)	4	2	5	5	1	1	4	3
Rhagionidae	Rhagionidae (DI)	–	–	5	4	–	–	1	1
<i>Simulium</i> sp.	Simuliidae (DI)	1	1	–	–	–	–	1	1
<i>Tipula</i> sp.	Tipulidae (DI)	4	1	4	4	1	1	3	2
<i>Paracladopelma</i> sp.*	Chironomini (DI)	4	3	2	2	1	1	4	3
<i>Paratendipes albimanus</i>	Chironomini (DI)	–	–	–	–	2	1	1	1
<i>Stictochironomus</i> sp.	Chironomini (DI)	4	2	4	4	4	3	4	2
<i>Prodiamesa olivacea</i>	Orthoclaadiinae (DI)	1	1	3	2	2	2	4	1
<i>Rheocricotopus</i> sp.	Orthoclaadiinae (DI)	–	–	1	1	–	–	2	1
<i>Cladotanytarsus</i> sp.*	Tanytarsini (DI)	4	4	3	3	2	2	4	4
<i>Lebertia rivulorum</i>	Lebertiidae (HYC)	4	1	–	–	–	–	4	3

differing background environmental conditions: for instance, mean current velocity in low-gradient lowland streams is naturally relatively low. This leads to a higher tolerance or resistance to low flows within the benthos,

because rheobionts are rare within the regional species pool. Another factor responsible for differences in response among streams might be their relative degradation by human activities. For instance, Dunbar *et al.*

Table 6 Taxa and their total abundance which were not recorded pre-manipulation and in the control reach after manipulation of the flow regime. Sampling occasions (stagnation/dry bed and pools) were $t = 1$, day 1/1; $t = 2$, day 8/4; $t = 3$, day 15/11; $t = 4$, day 22/18; $t = 5$, day 29/25 after manipulation. Current preference of the newcomers is derived from the freshwater ecology.info-database (Schmidt-Kloiber & Hering, 2012): LP, limnophil; LR, limno- to rheophil; RL, rheo- to limnophil; NA, not available

Taxon	Taxonomic group	Recorded at sampling occasion	Total abundance	Current preference
Stagnation				
<i>Lymnaea stagnalis</i>	Lymnaeidae (GA)	5	1	LR
<i>Planorbium corneus</i>	Planorbidae (GA)	5	1	LR
<i>Trocheta bykovskii</i>	Erpobdellidae (HI)	5	1	LR
<i>Arrenurus cylindricus</i>	Arrenuridae (HYC)	5	2	NA
<i>Hygrobatia longipalpis</i>	Hygrobatidae (HYC)	5	6	NA
<i>Cloeon dipterum</i>	Baetidae (EP)	5	1	LP
<i>Nepa cinerea</i>	Nepidae (HE)	5	1	LR
<i>Hesperocorixa sahlbergi</i>	Corixidae (HE)	5	1	LP
<i>Hydroporus palustris</i> adult	Dytiscidae (CO)	5	1	LR
<i>Agabus bipustulatus</i> adult	Dytiscidae (CO)	5	1	LR
<i>Gyrinus substriatus</i> adult	Gyrinidae (CO)	5	2	LR
<i>Procladius</i> sp.	Tanypodinae (DI)	2	2	NA
Pools				
<i>Agabus paludosus</i> adult	Dytiscidae (CO)	3,4	3	RL
<i>Culex pipiens/torrentium</i>	Culicidae (DI)	3,4,5	3270	LP
<i>Chironomus</i> sp.	Chironomini (DI)	4,5	27	NA
<i>Procladius</i> sp.	Tanypodinae (DI)	5	1	NA
Syrphidae	Syrphidae (DI)	3	3	NA
Dry streambed				
Ephydriidae	Ephydriidae (DI)	3,4	7	NA
Muscidae	Muscidae (DI)	4	2	NA

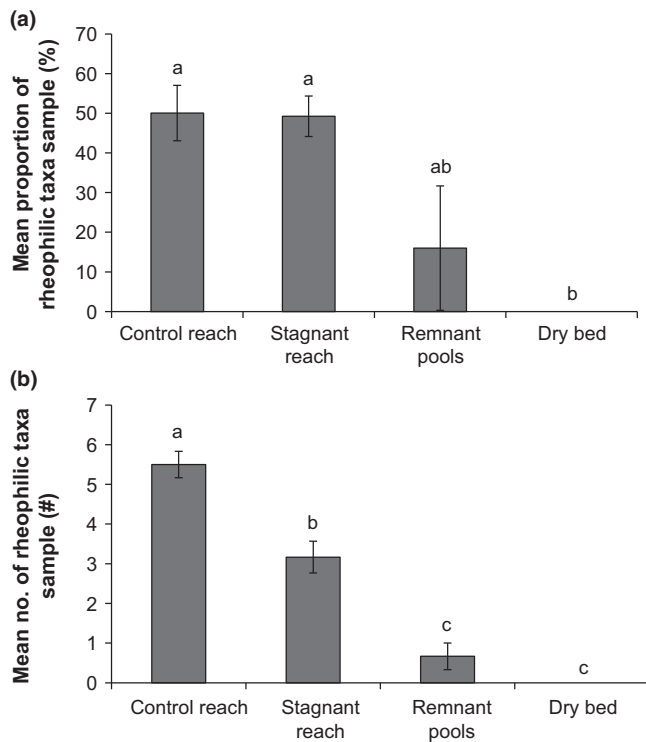


Fig. 6 Mean ($\pm$ SE) proportion (a) and number (b) of rheophilic taxa in the macroinvertebrate community at the end of the treatment period. Bars with different letters are significantly different (Mann–Whitney U-tests and Tukey *post hoc* comparisons, $P < 0.05$).

(2010) showed that physical habitat degradation leads to a stronger negative response of stream macroinvertebrates to low flows than in more intact streams of higher overall habitat quality.

Directly after flow ceased, an increase in taxon richness was observed. This increase could not be explained by a 'concentration effect', because water depth and width (and thus volume) in the stagnant reach remained similar and overall macroinvertebrate density did not change. A possibility may be that an increase in activity (e.g. resulting from the release from near-bed shear stress) affects the probability of capture. Another possibility might be that individuals of some taxa moved from the drying into the stagnant reach, via subsurface sediments (e.g. Hydracarina), the dam or even the riparian zone. Active overland dispersal is recorded in several groups of macroinvertebrates (Bilton, Freeland & Okamura, 2001), and we observed macroinvertebrates moving on the dry streambed after the surface water disappeared. This increase in richness was followed by a decline to pre-manipulation values 15 days after the onset of stagnation. The loss of rheophiles is insufficient to explain this temporal decrease in taxon richness; other potential explanations could be that pupation and/or

emergence was synchronised, physiological response to stagnation (Williams, 1987) or changes in predation rate (Boulton, 2003; Acuña *et al.*, 2005; Boulton & Lake, 2008).

Taxon richness in the control reach increased steadily from 8 days after manipulation, driven by background seasonal changes in the community (hatching of eggs, growth to a body size retained by the net). At the end of the experimental period, richness in the stagnant reach was similar to that in the control, masking one important difference in composition: newcomers contributed considerably to the richness of the stagnant reach at day 29. Many of those taxa were characteristic of standing water rather than streams. They may already have been present in the sheltered margins and moved into formerly unsuitable habitat as flow receded, although colonisation *de novo* is also likely. Many cases of the colonisation of lotic systems by lentic fauna after the loss of flow have been recorded (Smith & Pearson, 1987; Bond *et al.*, 2008; Boulton & Lake, 2008; Finn, Boulton & Chessman, 2009). Some of the taxa we observed are capable of flight (Coleoptera, Heteroptera, Ephemeroptera), or use such mobile colonists as a dispersal vector (Hydracarina). For the two species of lentic Gastropoda, *Lymnaea stagnalis* and *Planorbis corneus*, potential dispersal vectors at the study site were wild boar (*Sus scrofa*) and mallard (*Anas platyrhynchos*) (Van Leeuwen *et al.*, 2013).

Refuge value of small isolated pools

The response observed in the small isolated pools differed from that in the stagnant reach. Immediately after isolation, rises in taxon richness and overall density were observed in the pools, highlighting a concentration effect as the surface water receded. Some invertebrates actively or passively track the receding water and accumulate in the last remaining puddles (Extence, 1981; Dewson, James & Death, 2007b). It is important to note that our artificial pools differ in two ways from natural remnant pools and puddles. First, they cannot be colonised by movement through the sediment, but only from the surface. Second, there was a sharp delineation between the dry streambed and the ridges of the buckets, which differed from the gently sloping sediments normally found in natural pools. To what extent these differences affected the colonisation process is unknown, but the fact that more than sixty per cent of the total assemblage was recovered in the pools indicates that they were readily accessible.

Macroinvertebrate richness decreased rapidly in the week after the onset of streambed drying. Deteriorating

water quality, in our study indicated by hypoxia, potentially caused the rapid disappearance of most stream invertebrates, a common phenomenon in pools (Lake, 2003; Dewson *et al.*, 2007a). Because of the small volume of water in the pools, decomposition of plant-derived organic material that had accumulated on the bottom leads to rapid consumption of the dissolved oxygen available. Furthermore, the concentration of many organisms in a small volume of water could have resulted in strong biotic interactions (especially predation) and high mortality rates, in turn accelerating the depletion of oxygen from the water (Covich *et al.*, 2003; Acuña *et al.*, 2005; Boulton & Lake, 2008).

Loss of most of the macroinvertebrate assemblage indicated that small remnant pools were not sufficient to ensure survival for many taxa, even over a relatively short period. Instead, they were replaced by lentic taxa, such as *Culex* mosquitoes and *Chironomus* chironomids, which are both typical of the final stages of pool drying (Boulton & Lake, 2008). The contrasting macroinvertebrate assemblage response between the stagnant reach and the pools indicates that, at least in the lowlands of NW Europe, environmental conditions after cessation of flow are crucial in determining the persistence of stream invertebrates during water draw-down.

Refuge value of the sediments below the dry streambed

As in the small isolated pools, only a subset of the total assemblage was recorded in samples taken after the stream dried, and richness decreased in the weeks after manipulation. Nonetheless, 19 taxa were still alive after 11 days of streambed drying, and eight taxa survived up to 25 days buried in the surficial sediment. This observation supports the finding of Stubbington *et al.* (2009) that the bed could act as a streambed drying refuge for macroinvertebrates in stream systems dominated by fine-grained sediments. Sediment moisture content is an important determinant of macroinvertebrate survival in dry streambeds (Stanley *et al.*, 1994; Wickson, Chester & Robson, 2012; Stubbington & Datry, 2013). Although there was no free water present on the streambed during the study period, the sediment remained moist up to day 18 after the onset of streambed drying. Water loss was ameliorated by rain and by heavy shading from the forest canopy, which could have slowed the drying of the surficial sediment (Storey & Quinn, 2013). Also, the presence of (partly buried) leaf packs and woody debris could have facilitated macroinvertebrate survival, because of the moisture these microhabitats retain

((Paltridge *et al.*, 1997; Boulton & Lake, 2008; Chester & Robson, 2011).

Treatment effects

Due to the scale of our study and the types of treatment applied, we were unable to replicate the treatments in space, to a certain extent limiting the extrapolation of our findings to other systems. Although replication in time would likely have strengthened our results, the time span of the experiment makes such replication challenging. Another issue is that we do not know the extent to which the spatial configuration of the treatments affected our results. The sequentially decreasing levels of water from the control condition to the stagnation condition to the dry treatment may have affected invertebrate behaviour. We could have used separate channels for the different treatments, but the major strength of our approach was that we maximised the similarity of the macroinvertebrate assemblages using the same channel for all treatments. Despite these two limitations, we believe that our study provides new insights into temporary streams and, to our knowledge, these types of experimental manipulations of a 'natural' stream have not been carried out before.

In conclusion, our investigation demonstrated that the effect of a shift from perennial to intermittent flow on macroinvertebrates depends on the degree of habitat change. When flow ceases, but other environmental conditions remain within the range observed when flow is present, resistance is the main community response mechanism. With low flows and streambed drying as ramp disturbances, more intense and longer periods of such conditions (expected under some climate-change predications) will negatively affect community resistance. Shrinking of surface water to small isolated pools, or to a situation without surface water, not surprisingly had much larger effects on the richness and composition of the macroinvertebrate assemblage. Only a limited number of taxa survived in the pools or in the sediment beneath the dry streambed. In the pools, the presence of other stressors, such as hypoxia resulting from organic load, becomes important determinants of survival. Although temporary stream communities are characterised by high resilience (Fritz & Dodds, 2004), it is questionable if the same holds for assemblages in streams with perennial flow, of which part may have a relatively poor dispersal capacity (Sundermann, Stoll & Haase, 2011). With any increase in intensity and duration of low flows and streambed drying, an effect can be expected on larger spatial scales. Poor dispersers will be

particularly affected, because movements over longer distances within catchments will be required for recovery and persistence, and community resilience will decline. Such a decrease in community resilience and biodiversity could then have further consequences for ecosystem processes in streams (Ledger *et al.*, 2011).

Acknowledgments

We thank Dorine Dekkers, Luiz Moliner Cachazo, Mario Peiró Espí, and Paula Poveda Beneyto for their help in the field and processing of the macroinvertebrate samples. Natuurmonumenten Noordwest-Veluwe and Waterschap Vallei en Veluwe provided access to the study site. We would also like to thank Alan Hildrew and two anonymous reviewers for their very thorough review of the manuscript and suggestions for improvement. This study was funded by REFRESH nr. EU FP7 collaborative project 'Adaptive strategies to Mitigate the Impacts of Climate Change on European Freshwater Ecosystems' (REFRESH, grant agreement 244121, 2010–2014).

References

- Acuña V., Muñoz I., Giorgi A., Omella M., Sabater F. & Sabater S. (2005) Drought and postdrought recovery cycles in an intermittent Mediterranean stream: structural and functional aspects. *Journal of the North American Benthological Society*, **24**, 919–933.
- AQEM consortium (2002) Manual for the application of the AQEM system. A comprehensive method to assess European streams using benthic macroinvertebrates, developed for the purpose of the Water Framework Directive. http://www.aqem.de/ftp/aqem_manual.zip
- Bilton D.T., Freeland J.R. & Okamura B. (2001) Dispersal in freshwater invertebrates. *Annual Review of Ecology and Systematics*, **32**, 159–181.
- Bogan M.T. & Lytle D.A. (2011) Severe drought drives novel community trajectories in desert stream pools. *Freshwater Biology*, **56**, 2070–2081.
- Bond N.R., Lake P.S. & Arthington A.H. (2008) The impacts of drought on freshwater ecosystems: an Australian perspective. *Hydrobiologia*, **600**, 3–16.
- Boulton A.J. (2003) Parallels and contrasts in the effects of drought on stream macroinvertebrate assemblages. *Freshwater Biology*, **48**, 1173–1185.
- Boulton A.J. & Lake P.S. (2008) Effects of drought on stream insects and its ecological consequences. In: *Aquatic Insects: Challenges to Populations* (Eds Lancaster J. & Briers R.A.), pp. 81–102. CAB International, Wallingford.
- Carpenter S.R., Frost T.M., Heisey D. & Kratz T.K. (1989) Randomized intervention analysis and the interpretation of whole-ecosystem experiments. *Ecology*, **70**, 1142–1152.
- Chester E.T. & Robson B.J. (2011) Drought refuges, spatial scale and recolonisation by invertebrates in non-perennial streams. *Freshwater Biology*, **56**, 2094–2104.
- Covich A.P., Crowl T.A. & Scatena F.N. (2003) Effects of extreme low flows on freshwater shrimps in a perennial tropical stream. *Freshwater Biology*, **48**, 1199–1206.
- Dewson Z.S., James A.B.W. & Death R.G. (2007a) A review of the consequences of decreased flow for instream habitat and macroinvertebrates. *Journal of the North American Benthological Society*, **26**, 401–415.
- Dewson Z.S., James A.B.W. & Death R.G. (2007b) Invertebrate responses to short-term water abstraction in small New Zealand streams. *Freshwater Biology*, **52**, 357–369.
- Dunbar M.J., Warren M., Extence C., Baker L., Cadman D., Mould D.J. *et al.* (2010) Interaction between macroinvertebrates, discharge and physical habitat in upland rivers. *Aquatic Conservation: Marine and Freshwater Ecosystems*, **20**, S31–S44.
- Extence C.A. (1981) The effect of drought on benthic invertebrate communities in a lowland river. *Hydrobiologia*, **83**, 217–224.
- Finn M.A., Boulton A.J. & Chessman B.C. (2009) Ecological responses to artificial drought in two Australian rivers with differing water extraction. *Fundamental and Applied Limnology*, **175**, 231–248.
- Fritz K.M. & Dodds W.K. (2004) Resistance and resilience of macroinvertebrate assemblages to drying and flood in a tallgrass prairie stream. *Hydrobiologia*, **527**, 99–112.
- Hynes H.B.N. (1970) *The Ecology of Running Waters*. Liverpool University Press, Liverpool.
- KNMI (2014) KNMI'14: Climate Change scenarios for the 21st Century – A Netherlands perspective. Scientific Report WR2014-01, KNMI, De Bilt. www.climate-scenarios.nl
- Lake P.S. (2000) Disturbance, patchiness, and diversity in streams. *Journal of the North American Benthological Society*, **19**, 573–592.
- Lake P.S. (2003) Ecological effects of perturbation by drought in flowing waters. *Freshwater Biology*, **48**, 1161–1172.
- Lake P.S. (2011) *Drought and Aquatic Ecosystems: Effects and Responses*. John Wiley & Sons, Ltd, Chichester, UK.
- Ledger M., Edwards F.K., Brown L.E., Milner A.M. & Woodward G. (2011) Impact of simulated drought on ecosystem biomass production: an experimental test in stream mesocosms. *Global Change Biology*, **17**, 2288–2297.
- Lytle D.A. & Poff N.L. (2004) Adaptation to natural flow regimes. *Trends in Ecology and Evolution*, **19**, 94–100.
- McCune B. & Grace J.B. (2002) *Analysis of Ecological Communities*. MjM Software Design, Gleneden Beach, OR.
- Miller A.M. & Golladay S.W. (1996) Effects of spates and drying on macroinvertebrate assemblages of an intermittent and a perennial prairie stream. *Journal of the North American Benthological Society*, **15**, 670–689.

- Moth Iversen T., Wiberg-Larsen P., Birkholm Hansen S. & Hansen F.S. (1978) The effect of partial and total drought on the macroinvertebrate communities of three small Danish streams. *Hydrobiologia*, **60**, 235–242.
- Paltridge R.M., Dostine P.L., Humphrey C.L. & Boulton A.J. (1997) Macroinvertebrate recolonization after re-wetting of a tropical seasonally-flowing stream (Magela Creek, Northern Territory, Australia). *Marine and Freshwater Research*, **48**, 633–645.
- Schmedtje U. & Colling M. (1996) Ökologische Typisierung der aquatischen Makrofauna. Informations-berichte des Bayerischen Landesamtes für Wasserwirtschaft 4/96.
- Schmidt-Kloiber A. & Hering D. (eds.) (2012) www.freshwaterecology.info - the taxa and autecology database for freshwater organisms, version 5.0 (accessed on 6 May 2014).
- Smith R.E.W. & Pearson R.G. (1987) The macro-invertebrate communities of temporary pools in an intermittent stream in tropical Queensland. *Hydrobiologia*, **150**, 45–61.
- Stanley E.H., Buschman D.L., Boulton A.J., Grimm N.B. & Fisher S.G. (1994) Invertebrate resistance and resilience to intermittency in a desert stream. *American Midland Naturalist*, **131**, 288–300.
- Storey R.G. & Quinn J.M. (2013) Survival of aquatic invertebrates in dry bed sediments of intermittent streams: temperature tolerances and implications for riparian management. *Freshwater Science*, **32**, 250–266.
- Stubbington R. & Datry T. (2013) The macroinvertebrate seed-bank promotes community persistence in temporary rivers across climate zones. *Freshwater Biology*, **58**, 1202–1220.
- Stubbington R., Greenwood A.M., Wood P.J., Armitage P.D., Gunn J. & Robertson A.L. (2009) The response of perennial and temporary headwater stream invertebrate communities to hydrological extremes. *Hydrobiologia*, **630**, 299–312.
- Stubbington R., Wood P.J. & Reid I. (2011) Spatial variability in the hyporheic zone refugium of temporary streams. *Aquatic Sciences*, **73**, 499–511.
- Sundermann A., Stoll S. & Haase P. (2011) River restoration success depends on the species pool of the immediate surroundings. *Ecological Applications*, **21**, 1962–1971.
- Ter Braak C.J.F. & Šmilauer P. (2012) *Canoco Reference Manual and User's Guide: Software for Ordination (version 5.0)*. Microcomputer Power, Ithaca, New York.
- Van den Brink P.J. & Ter Braak C.J.F. (1999) Principal Response Curves: Analysis of time-dependent multivariate responses of a biological community to stress. *Environmental Toxicology and Chemistry*, **18**, 138–148.
- Van Leeuwen C.A., Huig N., Van der Velde G., Van Alen T.A., Wagemaker C.A.M., Sherman C.D.H. *et al.* (2013) How did this snail get here? Several dispersal vectors inferred for an aquatic invasive species. *Freshwater Biology*, **58**, 88–99.
- Verdonschot P.F.M., Hering D., Murphy J., Jähnig S., Rose N.L., Graf W. *et al.* (2010) Climate change and the hydrology and morphology of freshwater ecosystems. In: *Climate Change Impacts on Freshwater Ecosystems* (Eds Kernan M., Battarbee R.W. & Moss B.R.), pp. 65–83. Blackwell Publishing, Oxford.
- Wickson S., Chester E.T. & Robson B.J. (2012) Aestivation provides flexible mechanisms for survival of stream drying in a larval trichopteran (Leptoceridae). *Marine and Freshwater Research*, **63**, 821–826.
- Williams D.D. (1987) *The ecology of temporary waters*. Croom Helm, Kent.
- Wood P.J. & Petts G.E. (1999) The influence of drought on chalk stream macroinvertebrates. *Hydrological Processes*, **13**, 387–399.

(Manuscript accepted 11 September 2014)